Neutrino Mass Ratios in the Rope Hypothesis

The Brannen-Type Relation and Its Status in the Rope Framework

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Abstract

The neutrino sector is approached through a Brannen-type mass relation analogous to the charged-lepton Koide relation, differing by a characteristic offset. This paper reports the relation and its status honestly: it is a phenomenological fit of the same family as the lepton case, sharing the same conditional character, and its offset is the subject of a separate (corrected) analysis. It is a modeled relation, not a derivation. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants; the lepton-ratio caveat below is pinned in the test suite.

Scope of this paper

CLAIM: neutrino mass ratios are described by a Brannen-type relation in the same family as the lepton Koide relation.

NOT CLAIMED: a first-principles neutrino mass derivation. Phenomenological; status Conjecture, with the offset treated in a separate corrected note.

1. The Brannen-Type Relation

Brannen observed that the neutrino masses fit a Koide-like relation with a phase offset relative to the charged leptons. In the rope framework this offset is given a structural reading, developed — and corrected — in the companion notes on the neutrino offset.

2. Status

The relation reproduces the neutrino mass pattern at the same phenomenological level as the lepton Koide relation, and inherits the same caveat: it is a striking numerical fit whose first-principles status is not established. It is reported as a conjecture.

Status

Brannen-type neutrino relation — Modeled (Conjecture).

Offset origin — see corrected companion note; partially Open.

First-principles neutrino masses — Open.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.